Introduction to Circuits and Electronics
CIE 110

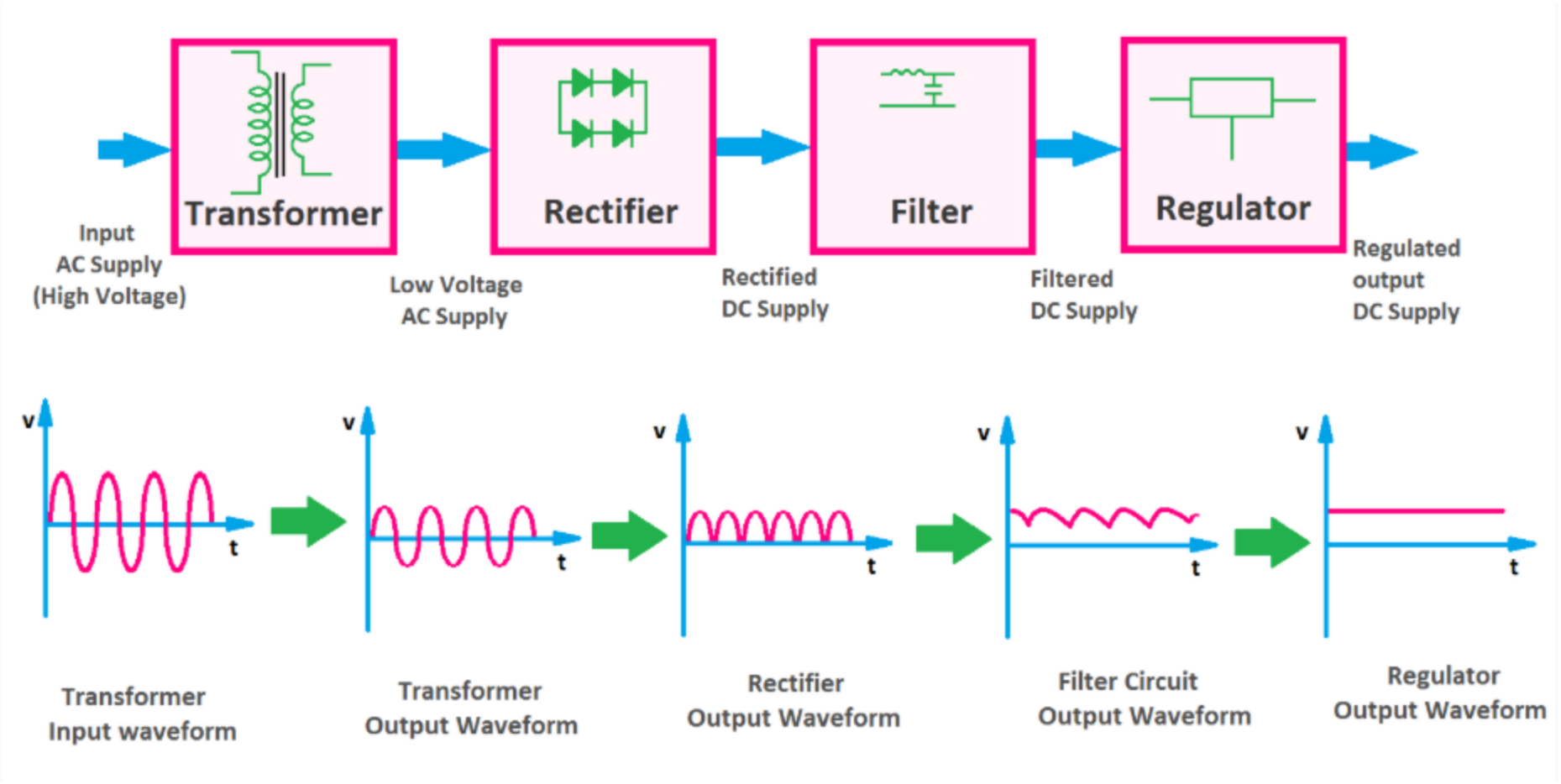
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Diode Applications

Diode Applications

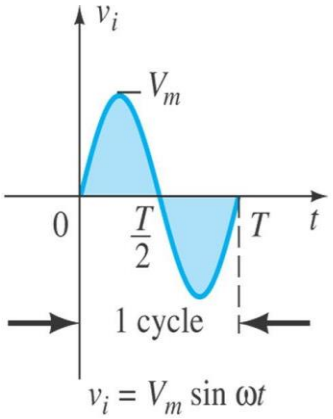
1. Rectifiers
2. Clippers or Limiters
3. Clampers
4. Voltage Multipliers

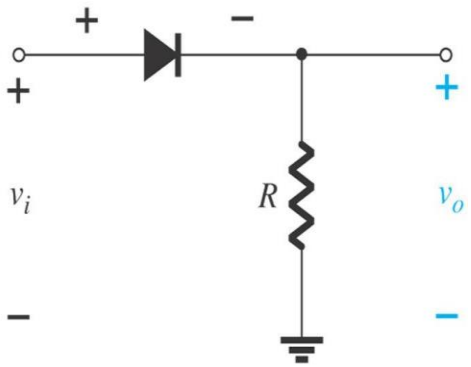
Our Target: Design a Power Supply



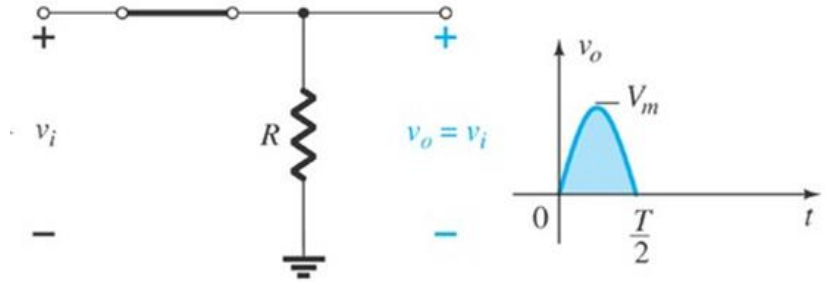
(1): Rectifiers

A) Half-Wave Rectification

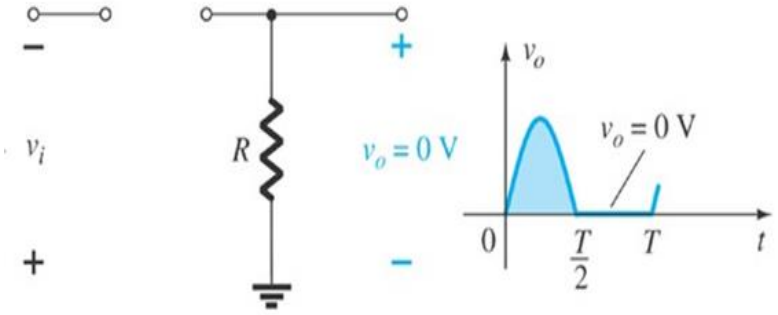




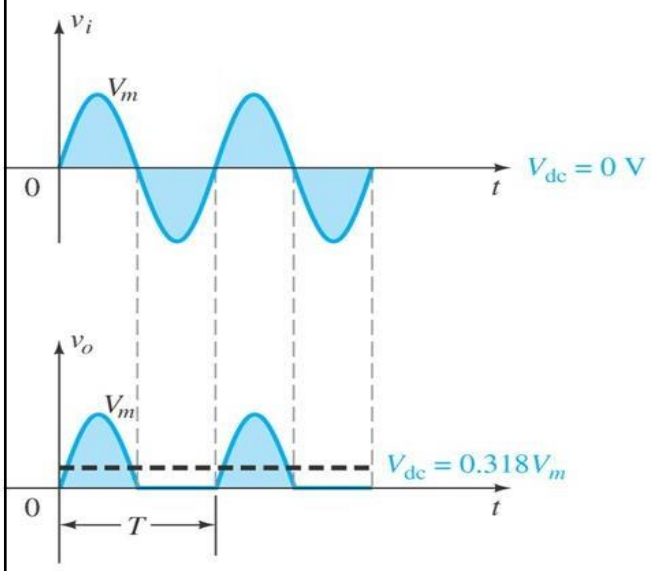
For t= 0 → T/2, the diode is on.
Assume ideal diode



For the period T/2 → T, the diode is off.
Diode is substituted with an open circuit.

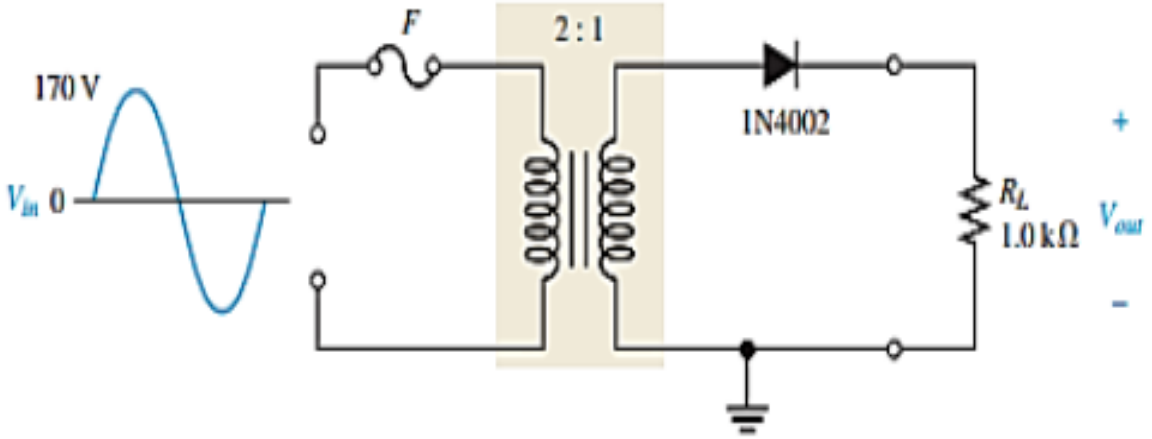


Parameters of Half-Wave Rectification

Half wave rectifier		▪ $V_{P(out)} = V_{P(\sec)} - 0.7\ V$ ▪ $V_{AVG} = \frac{V_{P(out)}}{\pi}$ ▪ $PIV\,(\text{peak inverse voltage}) = V_{P(\sec)}$ ▪ Frequency is constant ▪ If the frequency of input AC was 50 HZ then the expected frequency of the output AC from the half wave rectifier will be 50 HZ with no change in frequency.

Half wave rectifier with transformer coupled voltage

Determine the peak value of the output voltage for the shown rectifier if the turns ratio is 0.5



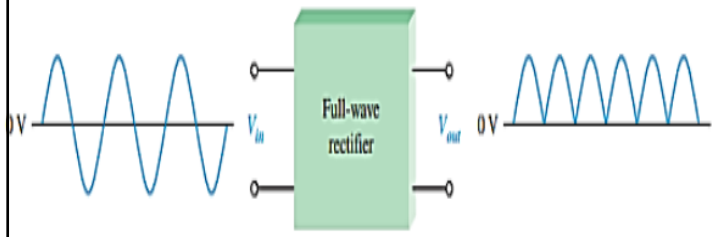
Solution:

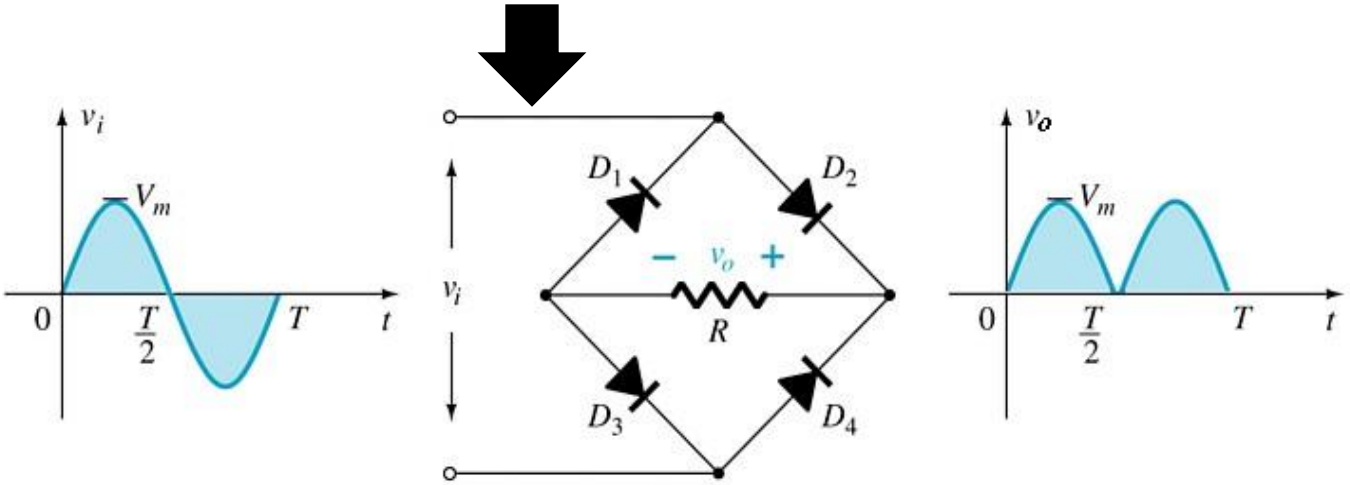
$V_{m(prim)} = V_{m(in)} = 170\ V$

The peak secondary voltage is: $V_{m(sec)} = nV_{m(pri)} = 0.5(170\ V) = 85\ V$

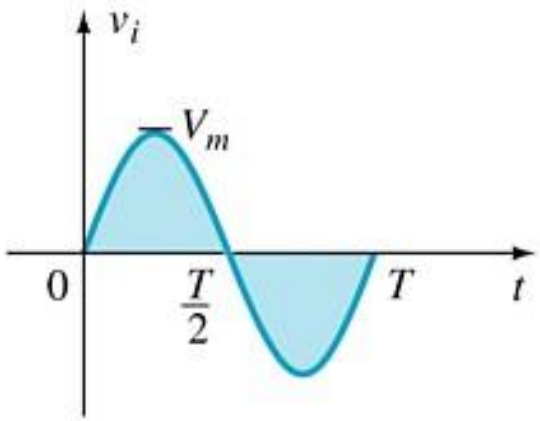
The rectified peak output voltage is $V_{m(out)} = V_{m(sec)} - 0.7\ V = 85\ V - 0.7\ V = 84.3\ V$

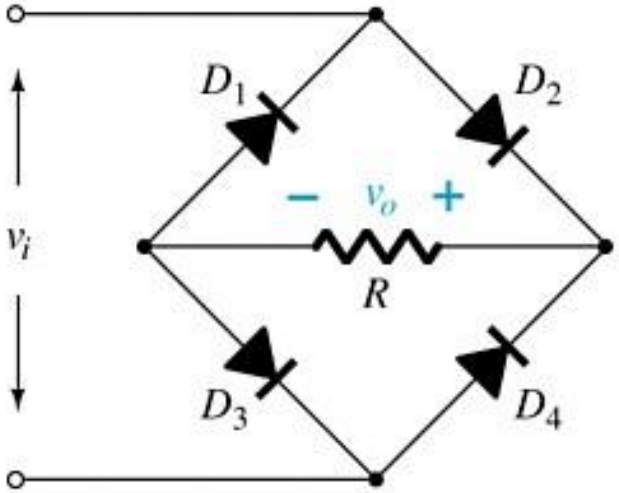
(B) Full-Wave Rectification

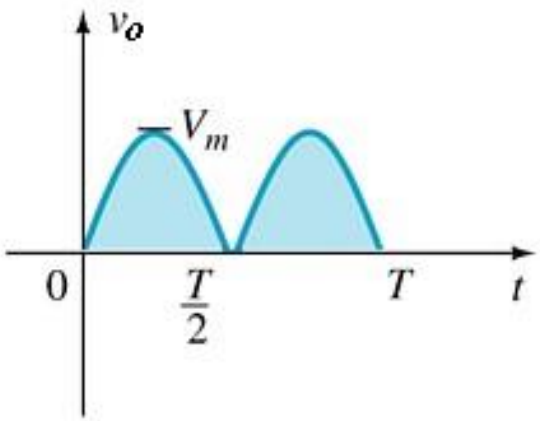
Bridge Full-Wave Rectifier		▪ $V_{P(out)} = V_{P(\text{sec})} - 1.4\ V$ ▪ $V_{AVG} = \frac{2V_{P(out)}}{\pi}$ ▪ $PIV = V_{P(\text{out})} + 0.7\ V$ ▪ Frequency is **doubled** ▪ If the frequency of input AC was 50 HZ then the expected frequency of the output AC from the Full wave rectifier will be 100 HZ.



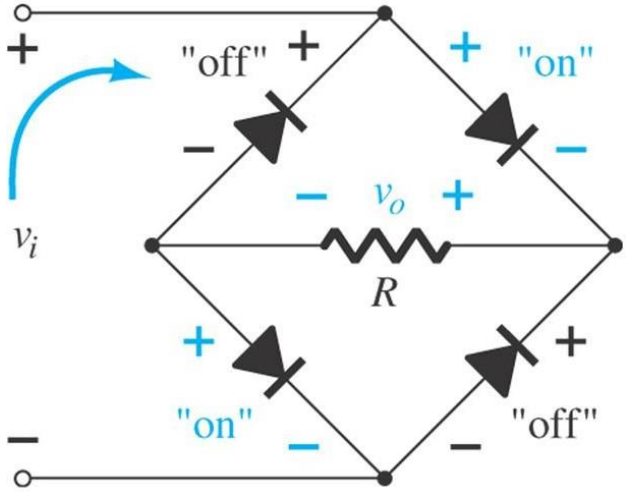
Full-Wave Rectification – Bridge Network



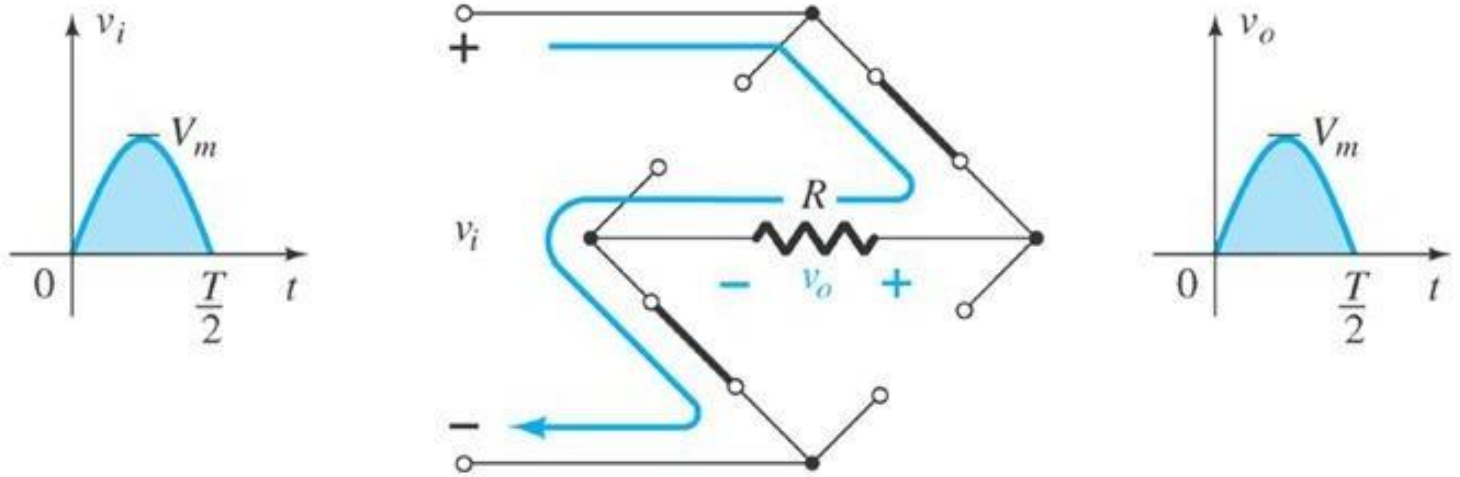




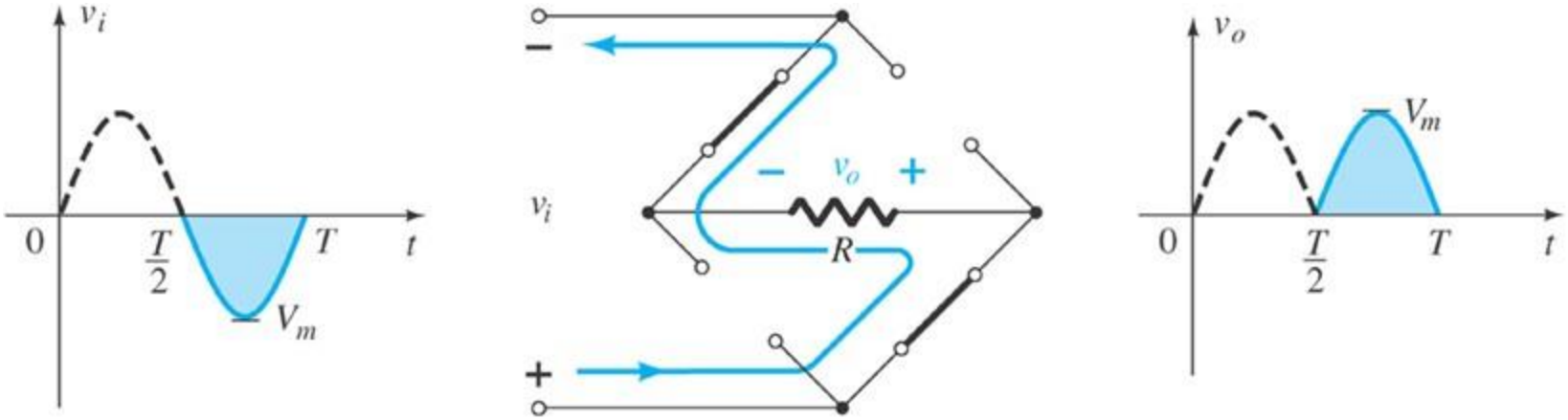
Network for the period 0 → T/2 of the input voltage v_i



Full-Wave Rectification – Bridge Network

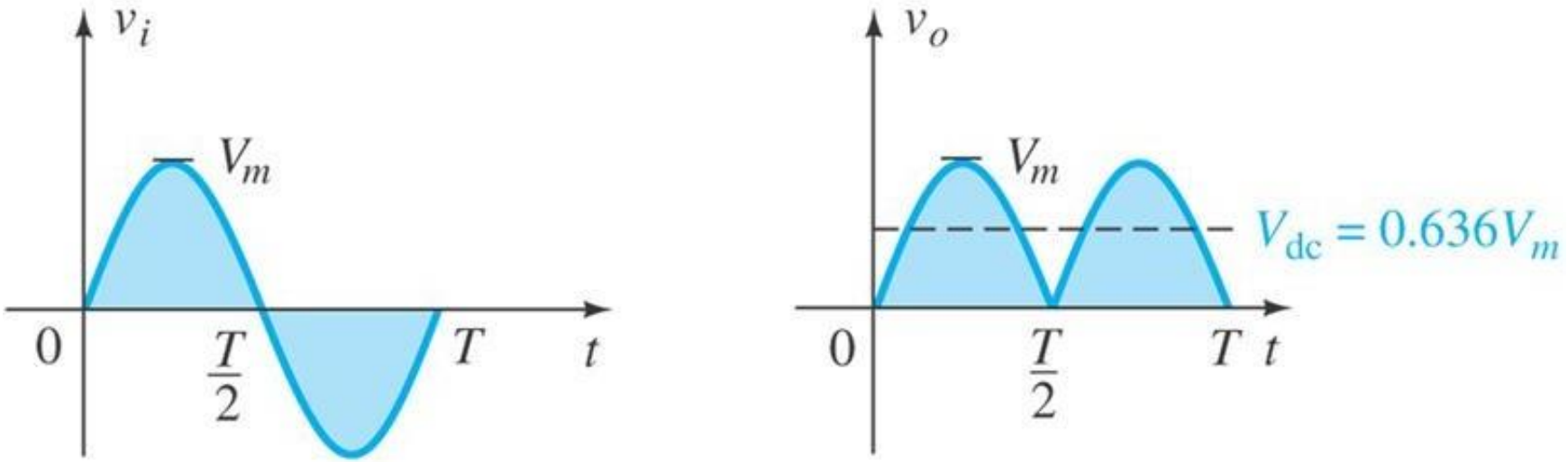


Conduction path for the positive region of v_i



Conduction path for the negative region of v_i

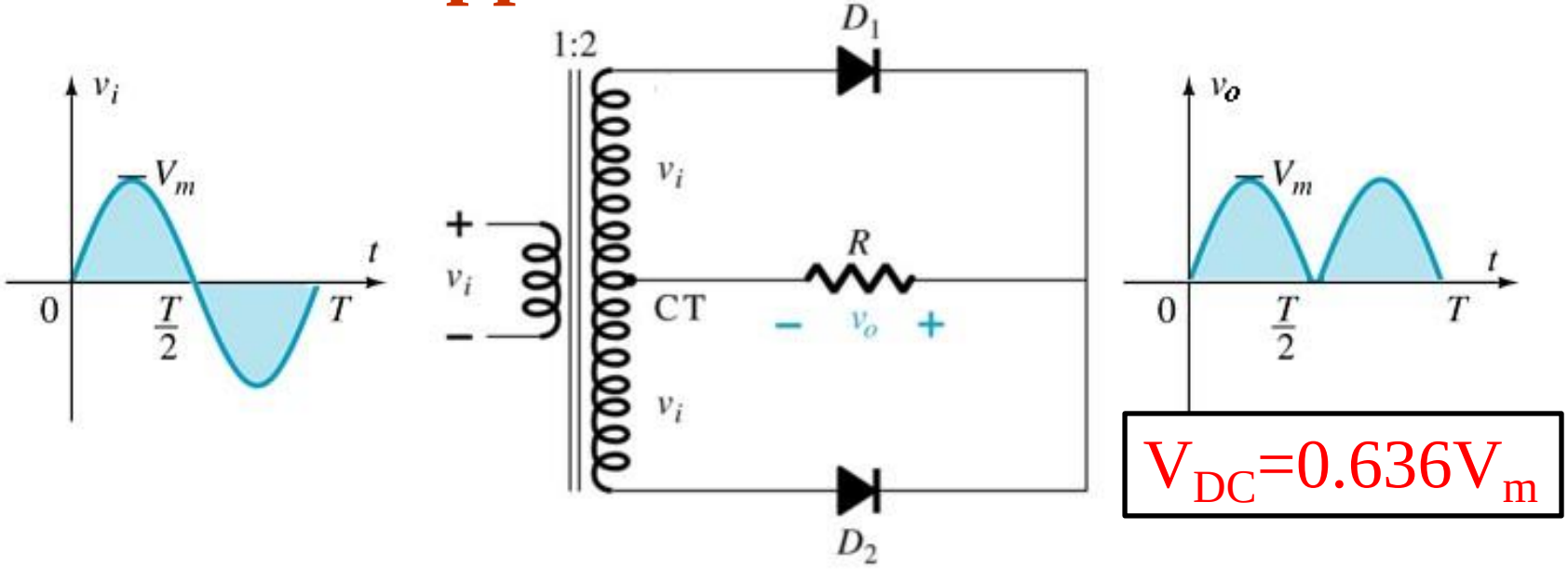
Full-Wave Rectification – Bridge Network



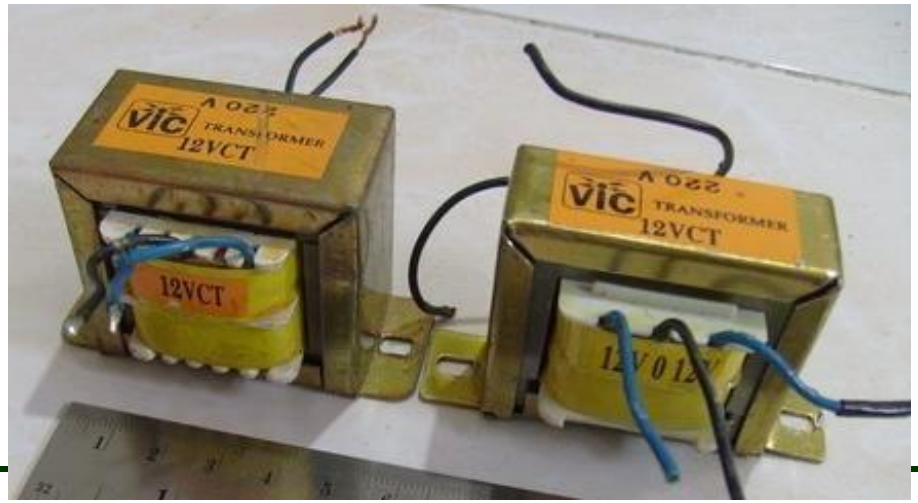
The DC level is now twice that of half wave rectifier=$2(0.318V_m)$

$$V_{DC}=0.636V_m$$

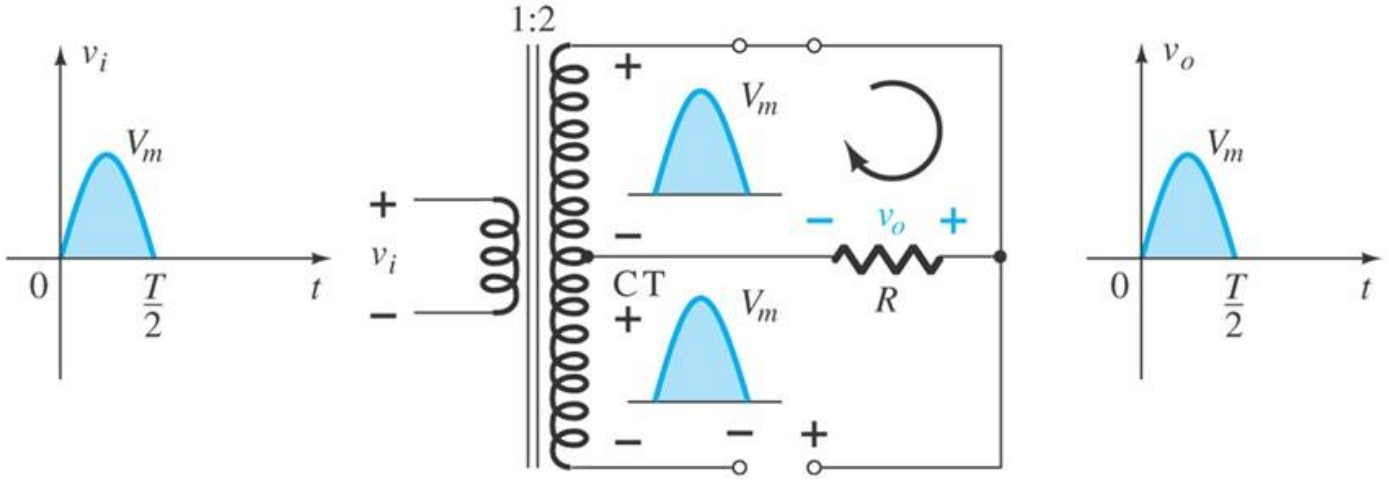
Full-Wave Rectification
Center Tapped Transformer Rectifier



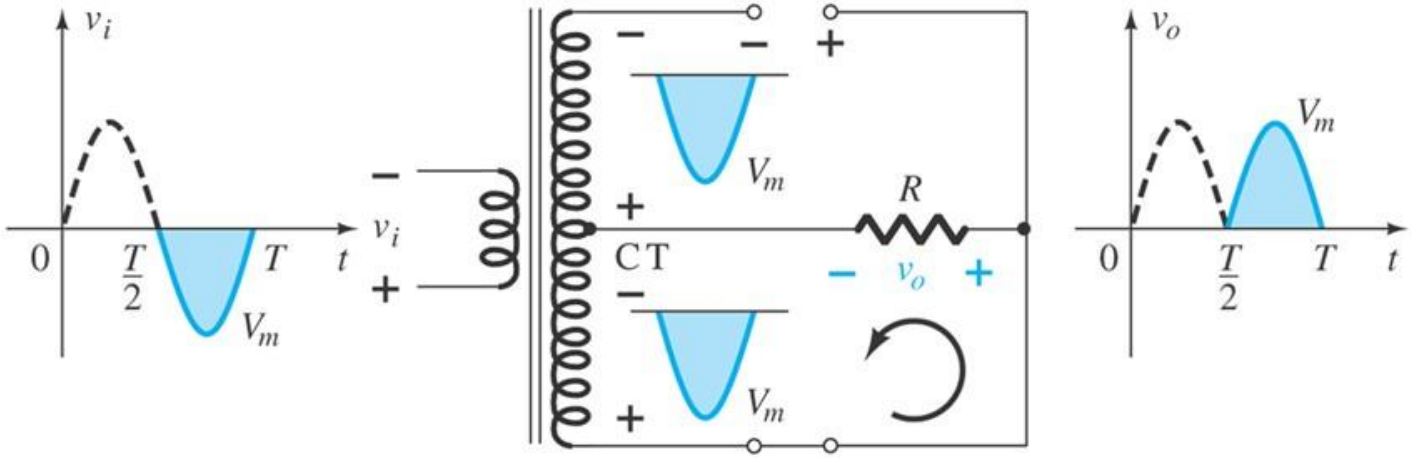
$V_{DC}=0.636V_m$



(C) Center Tapped Transformer Rectifier

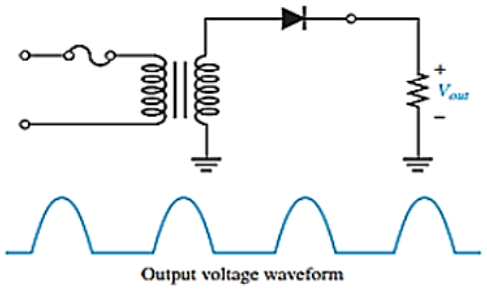
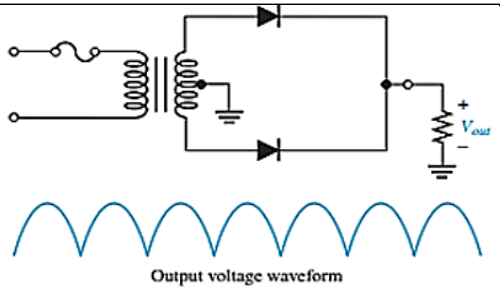
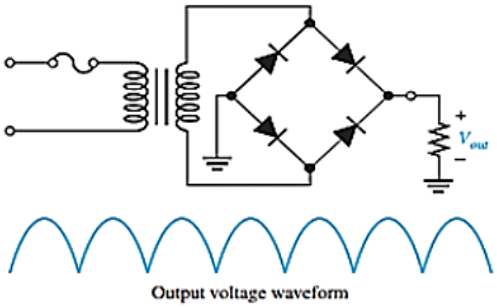


Network conditions for the positive region of v_i

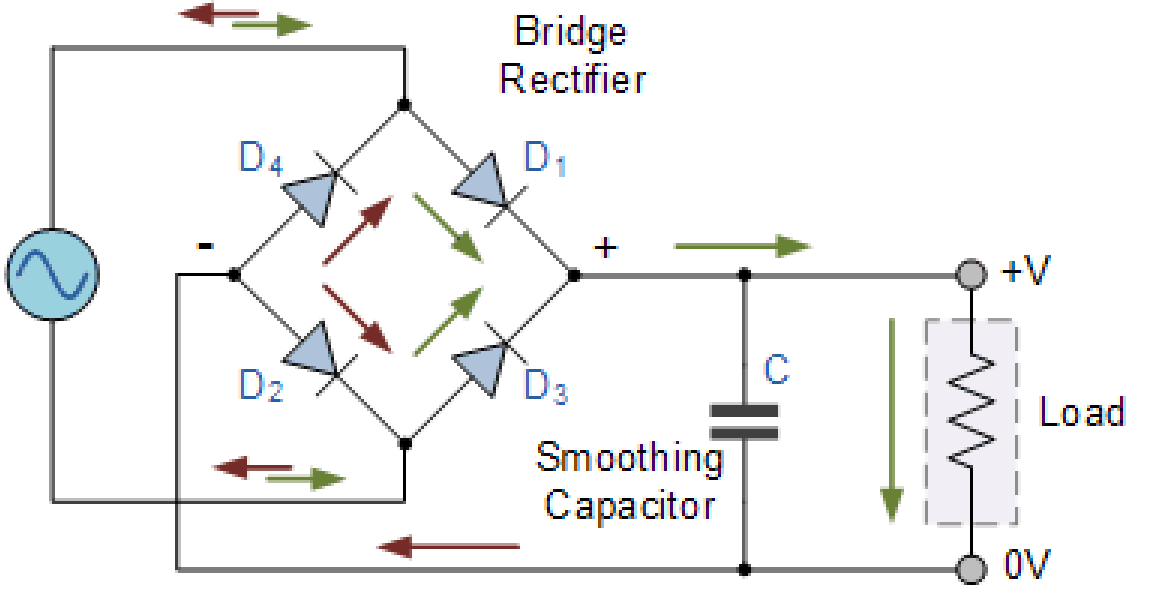


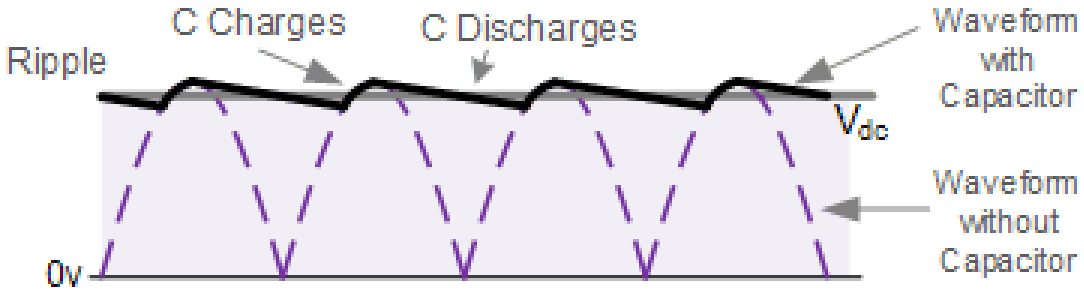
Network conditions for the negative region of v_i

Summary

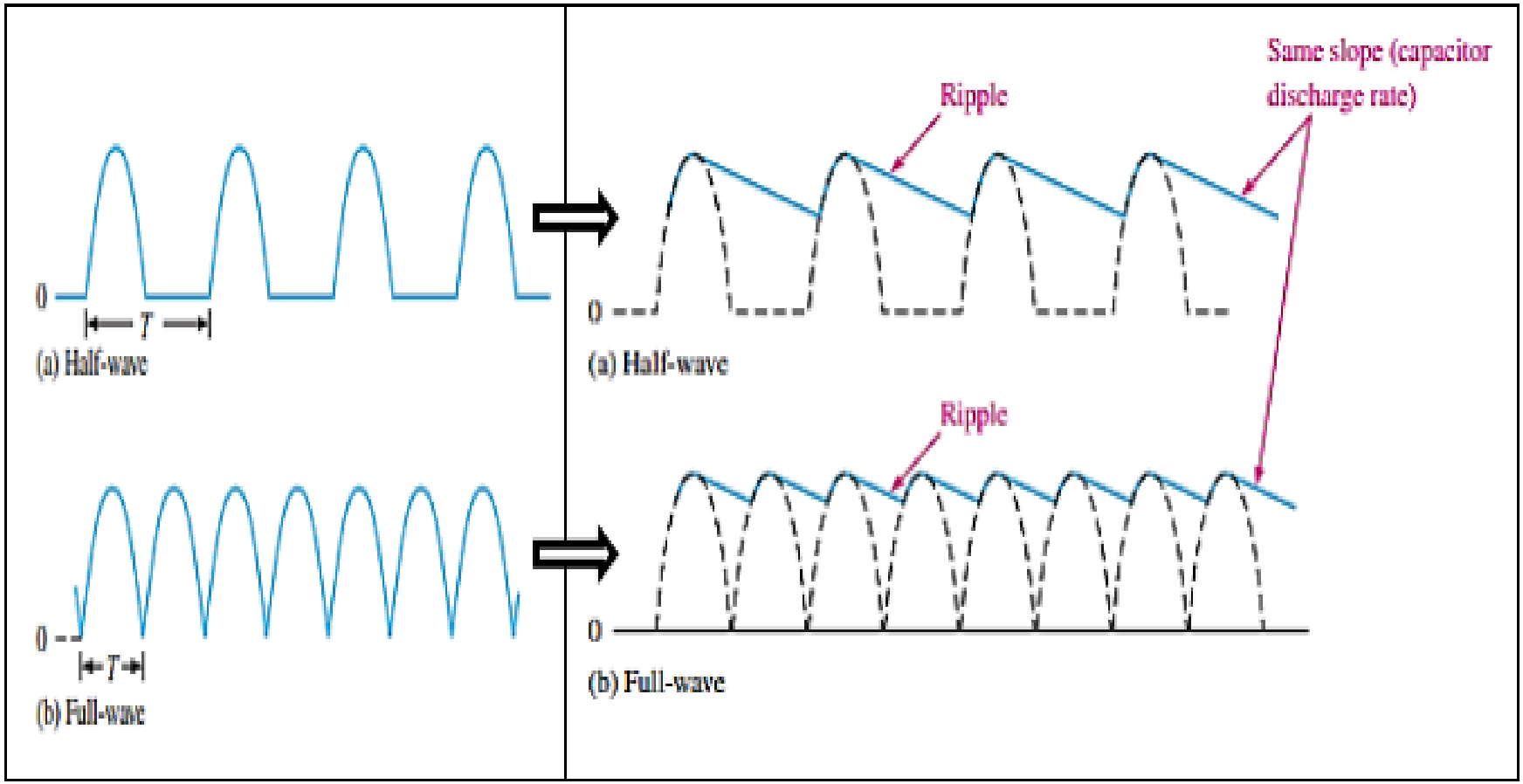
Half wave rectifier		$V_{P(out)} = V_{P(\sec)} - 0.7\ V$ $V_{AVG} = \frac{V_{P(out)}}{\pi}$ $PIV = V_{P(\sec)}$
Center-Tapped Full-Wave Rectifier		$V_{P(out)} = \frac{V_{P(\sec)}}{2} - 0.7\ V$ $V_{AVG} = \frac{2V_{P(out)}}{\pi}$ $PIV = 2V_{P(\text{out})} + 0.7\ V$
Bridge Full-Wave Rectifier		$V_{P(out)} = V_{P(\sec)} - 1.4\ V$ $V_{AVG} = \frac{2V_{P(out)}}{\pi}$ $PIV = V_{P(\text{out})} + 0.7\ V$

Full Wave Rectifier with Smoothing Capacitor (AC to DC Converter)





Resultant Output Waveform


0
T
(a) Half-wave
0
T
(b) Full-wave
Ripple
Same slope (capacitor discharge rate)
0
(a) Half-wave
Ripple
0
(b) Full-wave

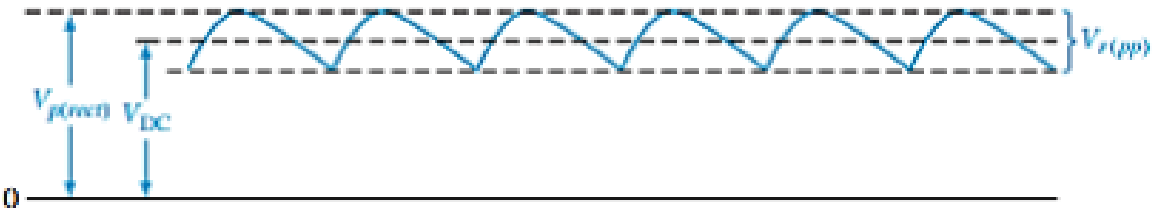
Ripple factor:

The ripple factor (r) is an indication of the effectiveness of the filter and is defined as:

$$r = \frac{V_{r(pp)}}{V_{DC}}$$

Where:

$V_{r(pp)}$:peak-to-peak ripple voltage



V_{DC} is the dc (average) value of the filter's output voltage

Important notes about ripple factor:

- The lower the ripple factor, the better the filter.
- The ripple factor can be lowered by increasing the value of the filter capacitor or increasing the load resistance
- For a full-wave rectifier with a capacitor-input filter:

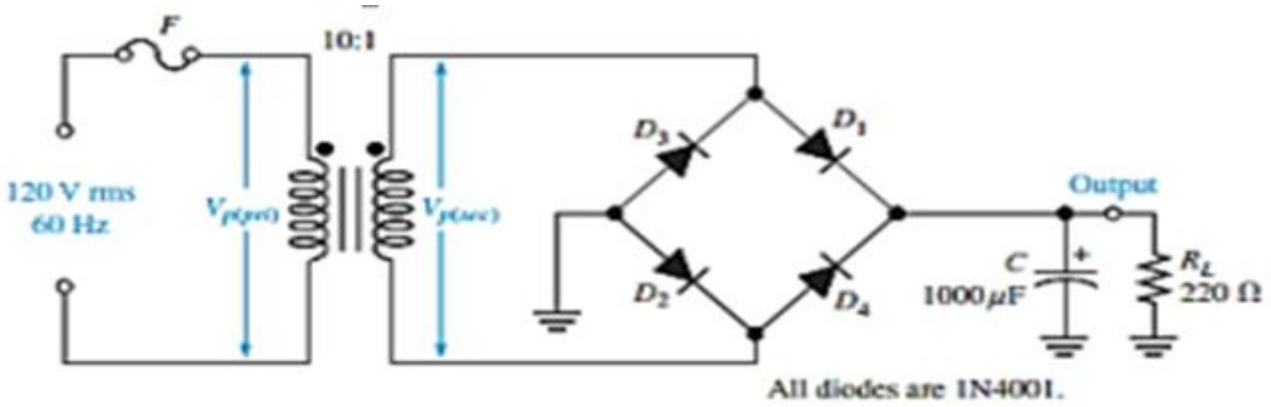
$$V_{r(pp)} = \left(\frac{1}{fR_LC}\right)V_{P(rect)}$$

$$V_{DC} = \left(1 - \frac{1}{2fR_LC}\right)V_{P(rect)}$$

Where: The variable $V_{p(rect)}$ is the unfiltered peak rectified voltage.

- If R_L or C increases, the ripple voltage decreases and the dc voltage increases.

Determine the ripple factor for the filtered bridge rectifier with a load as indicated in the shown figure.



$$V_{P\,(primary)} = 1.414 \times V_{rms} = 1.414 \times 120 = 170\ V$$
$$V_{P\,(secondary)} = nV_{P\,(primary)} = 0.1 \times 170 = 17\ V$$
$$V_{P\,(Rectified)} = V_{P\,(secondary)} - 1.4 = 17 - 1.4 = 15.6\ V$$

Full wave rectifier will **double** the frequency, accordingly:

$$V_{r(pp)} = \left(\frac{1}{fR_LC}\right)V_{P(rect)} = \left(\frac{1}{120 \times 220 \times 1000 \times 10^{-6}}\right) \times 15.6 = 0.591\ V$$
$$V_{DC} = \left(1 - \frac{1}{2fR_LC}\right)V_{P(rect)} = \left(1 - \frac{1}{240 \times 220 \times 1000 \times 10^{-6}}\right) \times 15.6 = 15.3\ V$$

Final ripple factor will be:

$$r = \frac{V_{r(pp)}}{V_{DC}} = 0.039 = 3.9\%$$